DATASCI 203 Final Project Report

Team Soccer - Summer 2026

Introduction

Football tournaments regularly showcase two strategies: teams that patiently build possession versus those that prioritize quick transitions, width, and runs behind the defense. The World Cup compresses these contrasting approaches into a single tournament. Yet despite decades of debate over which style is the best, it remains unclear which style is most effective at creating high-quality scoring chances.

Scoring chances can provide a better understanding of attacking performance compared to goals scored alone as goals are influenced by many factors unrelated to team play such as finishing ability, goalkeeper performance, and luck. Expected goals (xG) provides a more consistent way to measure attacking quality based on characteristics like shot location and chance type. Four tactical behaviors are especially likely to shape this quality. Possession-based build-up allows teams to progress the ball patiently and manipulate defensive shape before shooting, but it can also invite compact defending that limits central space. Attacking width stretches the opposition's back line horizontally, opening half-spaces and cutback lanes that tend to produce higher-quality shots than crosses into a crowded box. Vertical attacking movements, direct runs behind the defensive line, utilizes high defensive lines to create central, close-range changes, though they depend on defenses playing a high line in the first place. Finally, defensive aggression, how intensely a team presses and challenges for the ball out of possession, may influence how quickly a team regains possession and how many transition opportunities it creates. Each of these behaviors offers a distinct mechanism by which a team might play or fail to play dangerous shots.

This research is important for coaches and performance analysts because tactical decisions and playing styles can be changed and improved. Identifying which behaviors are strongly associated with higher xG can help teams make better decisions about training and match strategies, rather than findings that are interesting but outside their control. This study examines that relationship in the FIFA World Cup 2026: to what extent are tactical playing style indicators associated with a team's expected goals (xG)?

Research Question

To what extent are the four tactical concepts: possession-based build-up, attacking width, vertical attacking movements, and defensive aggression, playing style indicators associated with a team's expected goals (xG) during the FIFA World Cup 2026?

Description of the Data Source

The analysis uses the **FIFA World Cup 2026 Master Dataset** (Islam, 2026), a relational database distributed via Zendo (DOI: 10.5281/zenodo.21592427) and mirrored in Kaggle. The dataset was assembled by the creator through an automated mutli-source ingestion pipeline that collected the match data throughout the entire tournament, drawing from sources including fifa.com and sofacore.com. The data was normalized into 12 relational files covering match results, team statistics, player information, events, and other tournament-level information. The analysis uses the match-level team statistics data because it provides the variables necessary to measure tactical behavior and attacking output. For each match, two observations are created: one for each participating team. This structure allows the analysis to evaluate how a team's tactical characteristics during a specific match relate to its expected goals generated during that match.

The primary variables used are: *team expected goals* (`team_xg`), *corners* (`corners`), *possession percentage* (`possession_pct`), *offsides* (`offsides`), and *fouls* (`fouls`).

Data Wrangling

The analysis dataset combines two tables from the FIFA World Cup 2026 Master Dataset: `match_team_stats.csv`, which provides team-level tactical statistics (*possession, corners, offsides, fouls*), and `matches.csv`, which records *home_xg/away_xg* at the match level. Since the analysis requires xG at the team level, a Python pipeline reshapes the match-level xG into a team-level team_xg column before merging.

The pipeline is organized into three functions: `load_matches()` indexes matches.csv by *match_id*;`resolve_team_xg()` determines whether a given team played home or away and returns its corresponding xG and role; and `build_dataset()` orchestrates the merge and writes the result to `match_team_stats_with_team_xg.csv`. There are two validation checks, raising an error for any unmatched *match_id* or *team_id* to ensure join failures surface rather than pass silently. Because the script completed without error, all 208 team-match observations were successfully matched, and none required removal.

This output file is the single source of truth for all subsequent analysis; no intermediate dataframes are reused. It was then loaded into R, where the descriptive statistics, visualizations, and regression model were produced.

Operationalization

This study operationalizes tactical playing style using four match-level indicators from the FIFA World Cup 2026 Master Dataset, tested against team expected goals (team_xg) as the outcome. All 208 team-match observations were complete, with no observations removed. The dependent variable is team expected goals (team_xg), while the independent concept, tactical playing style, is represented by the variables corners (*corners*), Possession percentage (*possession_pct*), offsides (*offsides*), and fouls (*fouls*). Each of the chosen variables will represent one research concept. Team Expected Goals represent chance creation: measures the quality of scoring opportunities created during a match. *Possession percentage* represents build-up style: higher possession generally reflects a more patient, possession-based approach, though it can also reflect a losing team retaining low-risk possession while chasing the game. *Corners* represents attacking width: teams attacking through wide areas are more likely to generate corner kicks, though corners can also result from deflected shots or defensive clearances unrelated to deliberate wide play. *Offsides* represents vertical directness: frequent offsides suggest aggressive forward runs and attempts to play behind the defensive line, though offside counts depend jointly on a team's own directness and the opponent's defensive line height. *Fouls* represents defensive aggression: more fouls suggest higher defensive pressure and more aggressive on-ball challenges, though foul counts are also shaped by referee strictness, which varies by match. Alternative measures such as crosses or through balls could better represent attacking width and directness. However, these variables were not available in the dataset at the match level, making *possession percentage*, *corners*, *offsides*, and *fouls* the most appropriate proxies for the concepts examined in this study.

One or More Data Visualization(s)

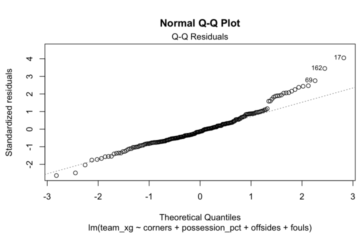


Figure 1: Normal QQ Plot

Figure 1 presents a Q-Q plot of the regression residuals to evaluate whether the assumptions of the linear regression model are reasonably satisfied. The residuals represent the difference between the observed expected goals and the values estimated by the model. If the points generally follow the reference line, this suggests that the residuals are approximately normally distributed and that the linear regression approach is appropriate. Deviations from the line, particularly at the extremes, indicate observations where the model performs less accurately. These diagnostic checks help assess the reliability of the regression estimates before interpreting the relationships between tactical variables and expected goals.

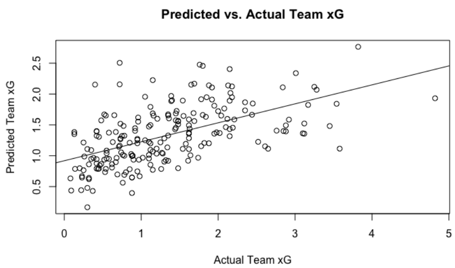


Figure 2: Predicted vs Actual Plot

```
Call:
lm(formula = team_xg ~ corners + possession_pct + offsides +
    fouls, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.78633 -0.46128 -0.09653  0.31971  2.88836

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.259995   0.301392   0.863  0.38935
corners         0.065857   0.021058   3.127  0.00202 **
possession_pct  0.023928   0.005289   4.524 1.03e-05 ***
offsides        0.012225   0.046078   0.265  0.79104
fouls          -0.037488   0.014597  -2.568  0.01094 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7189 on 203 degrees of freedom
Multiple R-squared:  0.3075, Adjusted R-squared:  0.2938
F-statistic: 22.53 on 4 and 203 DF,  p-value: 2.057e-15
```

Figure 3: Summary Table

Figure 2 compares observed team expected goals (xG) with the values estimated by the multiple linear regression model. Each point represents one team-match observation from the FIFA World Cup 2026 dataset. The diagonal line represents perfect agreement between the observed and estimated values. Observations closer to the line indicate that the model's estimates are closer to the actual expected goals values. This visualization provides an overview of how well possession, corners, offsides, and fouls collectively explain variation in team chance creation. While the model captures the general pattern of expected goals, deviations from the line indicate that other factors not included in the model may also influence chance creation.

Model Specification

To examine the relationship between tactical playing style and chance creation, we estimate a multiple linear regression model where team expected goals (xG) is the dependent variable. Expected goals is treated as a continuous outcome because it represents the estimated quality of scoring opportunities created during a match.

The model specification is: $team\_xg = \beta_0 + \beta_1 Possession + \beta_2 Corners + \beta_3 Offsides + \beta_4 Fouls + \epsilon$

where possession percentage represents build-up control, corners represent attacking width, and offsides represent vertical attacking behavior. These variables were selected because they represent tactical choices that coaches can directly influence during match preparation. No variable transformations were applied. The variables were maintained in their original units to preserve interpretability: possession remains a percentage, while corners and offsides remain match-level counts. This allows regression coefficients to be interpreted in football-relevant terms.

Regression Results: Tactical Factors Associated with Team Expected Goals

	Dependent variable: Team Expected Goals (xG)			
	(1)	(2)	(3)	(4)
Possession (%)	0.036***	0.027***	0.027***	0.024***
	(0.004)	(0.005)	(0.005)	(0.005)
Corners Created		0.060***	0.060***	0.066***
		(0.021)	(0.021)	(0.021)
Offsides			0.008	0.012
			(0.047)	(0.046)
Fouls				-0.037**
				(0.015)
Constant	-0.417*	-0.266	-0.273	0.260
	(0.213)	(0.216)	(0.221)	(0.301)
Observations	208	208	208	208
R2	0.257	0.285	0.285	0.307
Adjusted R2	0.253	0.278	0.274	0.294
Residual Std. Error	0.739 (df = 206)	0.727 (df = 205)	0.729 (df = 204)	0.719 (df = 203)
F Statistic	71.116*** (df = 1; 206)	40.829*** (df = 2; 205)	27.099*** (df = 3; 204)	22.531*** (df = 4; 203)

Figure 4: Table 1

Table 1 presents the multiple linear regression results; see Model Results and Interpretation below for detailed findings.

Model Assumptions

Multiple linear regression assumptions were evaluated using FIFA World Cup 2026 match data and diagnostic plots.

Linearity: The model assumes tactical variables have approximately linear relationships with expected goals (xG). While reasonable for identifying general trends, football relationships may be more complex than a linear model suggests.

Independence (IID): Observations represent individual team performances, but repeated matches from the same teams may create dependence due to shared tactics, players, and coaching. This may affect standard errors and significance tests.

Homoscedasticity: Residual plots were examined to ensure error variability remained consistent across predicted xG values. Violations could reduce the reliability of statistical tests.

Normality of Errors: The Q-Q plot indicated that residuals were approximately normally distributed, with minor deviations expected due to match variability.

Multicollinearity: Possession, corners, offsides and fouls may be related but represent distinct tactical dimensions. High correlation between variables could increase uncertainty in individual coefficient estimates.

Overall, the model provides descriptive evidence of tactical relationships with xG but should not be interpreted as causal due to factors such as opponent quality, player ability, and match context. The main limitation is potential dependence from repeated team observations.

Model Results and Interpretation

Regression Findings. Table 1 presents the multiple linear regression results examining the relationship between tactical indicators and team expected goals (xG). The overall model is statistically significant (F = 22.531, $p < 0.01$) and explains approximately 30.7% of the variation in team xG ($R^2 = 0.307$). This suggests that tactical behaviors contribute to chance creation, although other factors such as player quality, opponent strength, and match context also influence xG.

Possession Percentage: Possession has a positive and statistically significant relationship with xG (β = 0.024, $p < 0.01$). Holding other variables constant, teams with higher possession tend to create more scoring opportunities. However, possession alone does not guarantee attacking success, as its impact depends on how effectively teams use the ball. Corners Created: Corners have a positive and significant relationship with xG (β = 0.066, $p < 0.01$). This suggests that teams generating more corners often create greater attacking pressure and higher-quality chances, potentially through sustained possession in advanced areas and set-piece opportunities.

Offsides: Offsides have a positive but statistically insignificant relationship with xG (β = 0.012, $p > 0.05$). This indicates that attempting more runs behind the defense does not consistently translate into improved chance creation. The effectiveness of attacking movement may depend more on timing and execution than frequency.

Fouls: Fouls have a negative and statistically significant relationship with xG (β = -0.037, $p < 0.05$). Teams committing more fouls tend to produce lower expected goals, which may reflect reduced control of play, defensive pressure, or interruptions to attacking rhythm.

Overall, the regression results indicate that possession and corner generation are the strongest tactical indicators associated with higher expected goals, while offsides do not show a reliable relationship. Fouls appear negatively associated with attacking output. These findings suggest that effective chance creation is linked more to controlled possession and sustained attacking pressure than simply increased attacking activity.

Overall Effect

This study examined the relationship between tactical playing style and team expected goals in the FIFA World Cup 2026. Using multiple linear regression, we evaluated whether possession, corners, fouls, and offsides were associated with creating higher-quality scoring opportunities. The results indicate that possession and corner generation are positively associated with xG, while offsides do not show a statistically significant relationship, and fouls having a negative association to expected goals.

These findings suggest that effective chance creation is not simply a result of attacking more directly, but may depend on maintaining control of the match while creating repeated opportunities through attacking width. Although the model does not capture every factor influencing offensive performance, it provides evidence that certain tactical behaviors are more consistently associated with creating dangerous scoring chances. For coaches and performance analysts, these results highlight measurable aspects of team strategy that can be evaluated and adjusted during tournament preparation.

Appendix

Data Source & Provenance:

Islam, MD Mominul. "FIFA World Cup 2026 Dataset - Live & Daily Updated Stats." Kaggle, version 1.0.0, 2026, www.kaggle.com/datasets/mominullptr/fifa-world-cup-2026-dataset.

Data Prep & Access: Match statistics were merged and processed into match_team_stats_with_team_xg.csv to calculate team-level expected goals(xG) per match (N = 208)

Model Specifications Log:

We evaluated four incremental model specifications to examine how tactical plays can describe a team expected goals (xG)

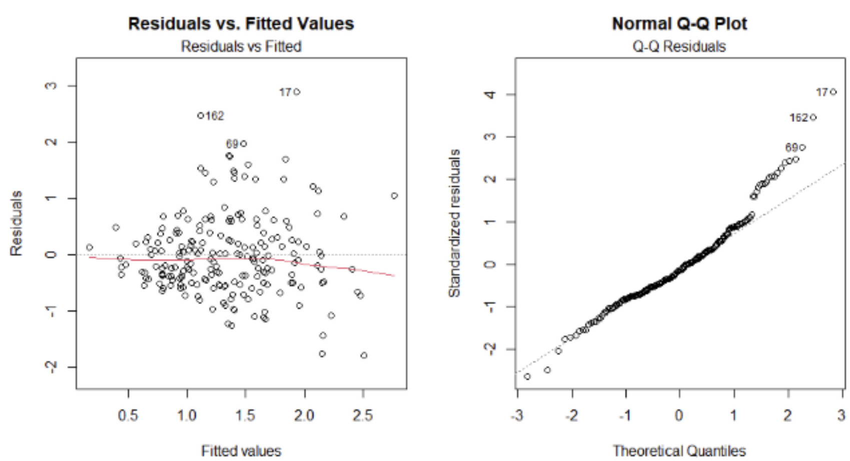
Model 1 (team_xg ~ possession_pct): Simple baseline model establishing the bivariate linear relationship between possession percentage and expected goals.

Model 2 (+ corners): Added set piece creation. This significantly improved model fit (R^2 = 0.285), depicting that attacking pressure via corners adds a descriptive value beyond possession.

Model 3 (+ offsides): Introduced offsides as a proxy for high attacking line risk/chance. It was found that offsides were not statistically significant (p = 0.791), indicating that aggressive forward runs behind opposing defensive lines alone does not describe chance quality.

Model 4 (+ fouls): Added tactical fouls committed showed a statistically negative association (Beta = -0.037, P= 0.011), suggesting defensive disruptions correlate with lower team expected goals.

Model Diagnostics and Residual Analysis:



Residuals vs Fitted Values Plot: The red smooth trendline remains relatively flat and centered near 0 across most fitted values from 0.5 to 2.0. This confirms that the linearity assumption holds well and variance remains relatively constant across predicted ranges.

Normal Q-Q Plot: Standardized residuals are seen close to the theoretical normal line through the majority of the data from quantiles from -2 to +1, supporting normality of errors assumption.

Outlier & Heavy Right Tail: The upper right quadrant shows right tail outlier driven by high-scoring matches (observation 69,162,17). This right skew is common in sports analytics, as teams occasionally generate unusually high chance/scoring games.